Perceived “Resource Dependence” and Canadian Public Universities

(My Attempt at) Building a Fine-Tuned Classifier via LLM Encoder

Sara Murphy, Computational Text Analysis, April 13, 2026

Overview

- **Goal:**
  - Detect presence of perceived "resource dependence" language for public universities
  - Detect intensity of perceived "resource dependence" language – low vs high
- **Methods:**
  - Built Fine Tuned Classifier LLM Encoder (0,1) – perceived resource dependence (no or yes)
  - Built Fine Tuned Classifier LLM Encoder (0,1) – intensity of perceived resource dependence (low or high)
- **Future Application:**
  - Relationship between perceived resource dependence and financial performance (presence, intensity)
  - Predict financial performance in provinces outside Ontario (e.g. Nova Scotia)
  - Link perceived resource dependence to leadership background and frequency of change
- **Challenges/Limitations:**
  - Sample size
  - Variation in documents across universities
  - Conceptual ambiguity and human validation
  - Moving to application stage

Context:

Public Universities in Ontario (Post 2019)

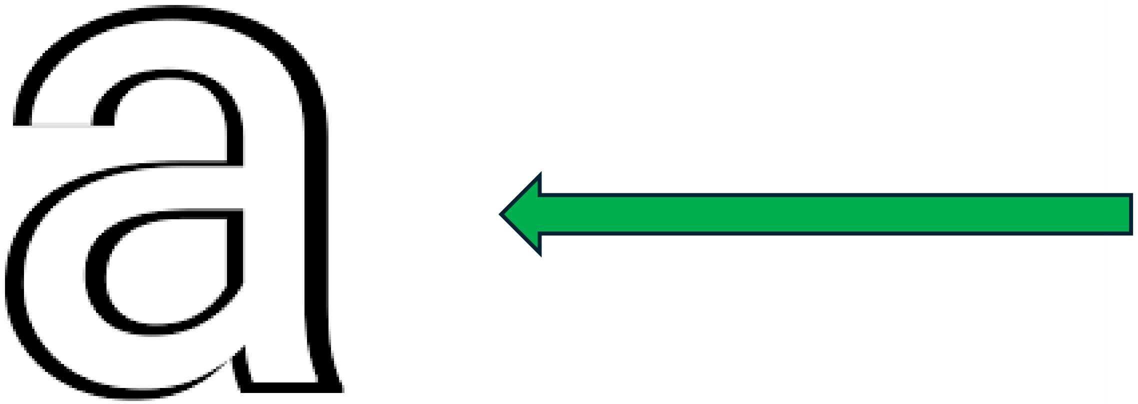

($63 million)
-9%
“Queen’s could cease to exist”

1/2

(~$300M)

(~$600M)


I can't
do it



Construct:

Resource Dependence

Resource Dependence

- **Power & Resource Dependence,** Emerson (1962)
  - "Power is the inverse of one's dependency on the other in a relationship"
  - "Dependence = motivational investment in the same goals/rewards, inverse of availability of alternative means of achieving goals outside of the relationship"

- **Resource Dependence is not an objective concept**
  - Yet typically studied in concrete terms (finances, resource flows)
- **There is a subjective, perceived or performative reality**
- **No dictionaries or LLMs assess perceptions of RD via text analysis**

Research Questions:

What a Fine-Tuned Classifier Could Help Answer

Research Questions

1) How does perceived resource dependence moderate the financial performance impact of the 2019 policy shock?

2) How does perceived resource dependence change in response to the 2019 policy shock ?

3) How does leadership succession cause change in perceived resource dependence pre and post the 2019 policy shock?

Data Collection, Processing & Sampling

Data: Original Plan

- **Operating Budgets**
  - Ontario x 20 universities
  - British Columbia & Nova Scotia X 18 universities
  - 2014-2025

- **Unit of Analysis**
  - **Introduction Section**
    - “Setting the Context”, “Introduction”, “Executive Summary”, “Changing Financial Landscape”
    - 1-5 pages of document
  - **Paragraph Level**
    - 5 – 45 paragraphs per introduction section
  - **No text pre-processing**
  - **Less than 288 tokens per paragraph**

- **Size of Total Corpus**
  - 38 universities X 12 years X avg 12 paragraphs = 5,472 paragraphs

Building My Corpora

document	ID	text	class_binary	class_intensity	order	total para	university	province	year	treatment group	treatment period	pr
queens_2017	queens_2017_1		1	1	1	9	Queen's University	ON	2017	1	0	
queens_2017	queens_2017_2		1	1	2	9	Queen's University	ON	2017	1	0	
queens_2017	queens_2017_3		1	0	3	9	Queen's University	ON	2017	1	0	
queens_2017	queens_2017_6		1	0	6	9	Queen's University	ON	2017	1	0	
queens_2017	queens_2017_8		1	1	8	9	Queen's University	ON	2017	1	0	
queens_2017	queens_2017_9		1	0	9	9	Queen's University	ON	2017	1	0	
queens_2018	queens_2018_1		1	1	1	9	Queen's University	ON	2018	1	0	
queens_2018	queens_2018_2		1	1	2	9	Queen's University	ON	2018	1	0	
queens_2018	queens_2018_6		1	0	6	9	Queen's University	ON	2018	1	0	
queens_2018	queens_2018_7		1	1	7	9	Queen's University	ON	2018	1	0	
queens_2018	queens_2018_9		1	0	9	9	Queen's University	ON	2018	1	0	
queens_2019	queens_2019_1		1	1	1	7	Queen's University	ON	2019	1	1	
queens_2019	queens_2019_2		1	1	2	7	Queen's University	ON	2019	1	1	
queens_2019	queens_2019_5		1	0	5	7	Queen's University	ON	2019	1	1	
queens_2019	queens_2019_6		1	0	6	7	Queen's University	ON	2019	1	1	
queens_2019	queens_2019_7		1	0	7	7	Queen's University	ON	2019	1	1	
queens_2020	queens_2020_5		1	1	5	10	Queen's University	ON	2020	1	1	
queens_2020	queens_2020_6		1	1	6	10	Queen's University	ON	2020	1	1	
queens_2020	queens_2020_8		1	0	8	10	Queen's University	ON	2020	1	1	
queens_2020	queens_2020_9		1	0	9	10	Queen's University	ON	2020	1	1	
queens_2020	queens_2020_10		1	0	10	10	Queen's University	ON	2020	1	1	
queens_2021	queens_2021_1		1	1	1	6	Queen's University	ON	2021	1	1	
queens_2021	queens_2021_2		1	1	2	6	Queen's University	ON	2021	1	1	
queens_2021	queens_2021_4		1	1	4	6	Queen's University	ON	2021	1	1	
queens_2021	queens_2021_5		1	0	5	6	Queen's University	ON	2021	1	1	
queens_2022	queens_2022_1		1	1	1	7	Queen's University	ON	2022	1	1	
queens_2022	queens_2022_2		1	1	2	7	Queen's University	ON	2022	1	1	
queens_2022	queens_2022_3		1	[illegible]	[illegible]	7	Queen's University	ON	2022	1	1	
queens_2022	queens_2022_4		1	[illegible]	[illegible]	7	Queen's University	ON	2022	1	1	
queens_2022	queens_2022_5		1	0	5	7	Queen's University	ON	2022	1	1	
queens_2022	queens_2022_6		1	0	6	7	Queen's University	ON	2022	1	1	
queens_2022	queens_2022_7		1	1	7	7	Queen's University	ON	2022	1	1	
queens_2023	queens_2023_1		1	1	1	8	Queen's University	ON	2023	1	1	
queens_2023	queens_2023_2		1	1	2	8	Queen's University	ON	2023	1	1	
queens_2023	queens_2023_3		1	0	3	8	Queen's University	ON	2023	1	1	

Variables & Codebook: Fine Tuner (1)

Class_Binary

- 0 = No Perceived Resource Dependence
- 1 = Perceived Resource Dependence

- **Label 0 = ANY paragraph that describes policy, funding, or risk WITHOUT evaluative language about constraint or opportunity**
- **Key Words and Phrases:**
  - References to the global economy , Covid-19 pandemic
  - References to budgeting process, descriptive budget information only
  - general references to policies or strategic mandate agreements without any adjectives or descriptive words implying dependence

- **Label 1 = ANY paragraph that clearly emphasizes at least one of:**
  - alternative revenue source/generation/diversification
  - agency (“we will”, “we can”) framing of government policy
  - opportunity framing of government policy or funding
  - Challenge or constraint framing of government policy or funding

Variables & Codebook: Fine Tuner (2)

Class_Intensity

- 0 = Low Perceived Resource Dependence
- 1 = High Perceived Resource Dependence

- **Label 0 - Key word/phrase indicators (low perceived resource dependence)**
  - *Policy is a welcomed change*
  - *Need to diversify revenue*
  - *Alternative revenue generation*
  - *Providing some flexibility*
  - *Committed to balanced budgets*
  - *Flexibility in the form of a contingency fund*
  - *Push for advocacy*
- **Label 1 = ANY paragraph that explicitly describes government policy or funding in terms of constraint, with no mention of agency or opportunity or alternative revenue sources**
- **Key word/phrase indicators of Label 1 (high resource dependence language)**
  - *being in a vulnerable position*
  - *provincial grant revenue, although presenting with some certainty, is still financially limiting for the university*
  - *limited financial flexibility*
  - *the tuition framework continues to restrict flexibility for the university*
  - *restricted flexibility due to domestic tuition freeze*
  - *limited revenue growth opportunities*

Queen's UNIVERSITY

UNIVERSITY OF TORONTO

Strategic Management Agreement (SMA)

"**at-risk** funding"

"the funding in the **performance-based** envelope will be at-risk **assuming the targets in each metric are continually met**"

"provides **predictability** and **stability**"

"this is a **welcomed change**"

"The **differentiation policy** framework re-confirms the university's **leadership role in research and innovation**"

"the university's long-term **advocacy** for differentiation"

"the government decides the metrics, **but the university sets the weights**, and is measured **against their own performance** not other institutions"

Binary Classifier (Perceived RD)

Phase 1

- **Queen's University**
  - 2017-2025
- **University of Toronto**
  - 2017-2025

- **Total observations:**
  - 327
  - Class 0 – No = 184
  - Class 1 – Yes = 143

Add Sample

Phase 4

- **Total observations:**
  - 644
  - Class 0 – No = 307
  - Class 1 – Yes = 337

Binary Classifier (Perceived RD Intensity)

Phase 2

- **Queen's University**
- **University of Toronto**
- **Carleton University**
- **University of Windsor**
  - Various Years
  - Post 2019

- **Total observations:**
  - 236
  - Class 0 – Low = 154
  - Class 1 – High = 82

Revise Codes

Phase 3

- **Queen's University**
- **University of Toronto**
- **Carleton University**
- **University of Windsor**
- **York University**
- **Nipissing University**
- **Lakehead University**
- **Wilfrid Laurier University**
- **University of Waterloo**
- **Western University**
- **McMaster University**
  - Various Years
  - Post 2019

- **Total observations:**
  - 337
  - Class 0 – Low = 214
  - Class 1 – High = 123

Add Sample

Binary Classifier (Perceived RD)

Phase 1

V14. xlsx

	Model	LR	Epochs	Batch Size	Mean Acc	SD Acc	Mean Weighted F1	SD Weighted F1	Mean F1_0	SD F1_0	Mean F1_1	SD F1_1
0	roberta-base	0.00002	4	16	0.636061	0.082057	0.547946	0.151291	0.29978	0.315747	0.74142	0.029538

RoBERTa, Google Colab
Stratified K Fold
No Class Weight Loss
No Weighted Sampling

Binary Classifier (Perceived RD)

9 collapsed folds

Phase 1a

	fold	Train Loss	Val Loss	Accuracy	Weighted F1	F1_0	F1_1
0	1	0.698057	0.671035	0.560606	0.402766	0.000000	0.718447
1	1	0.664254	0.677310	0.560606	0.402766	0.000000	0.718447
2	1	0.568641	0.619747	0.636364	0.566919	0.333333	0.750000
3	1	0.505898	0.623363	0.681818	0.659631	0.533333	0.758621
4	2	0.689950	0.684404	0.560606	0.402766	0.000000	0.718447
5	2	0.681073	0.624862	0.590909	0.467437	0.129032	0.732673
6	2	0.547687	0.453209	0.757576	0.752681	0.692308	0.800000
7	2	0.396852	0.373532	0.772727	0.770956	0.727273	0.805195
8	3	0.685684	0.656347	0.569231	0.412971	0.000000	0.725490
9	3	0.671715	0.705432	0.569231	0.412971	0.000000	0.725490
10	3	0.621074	0.501803	0.738462	0.739704	0.721311	0.753623
11	3	0.491929	0.503208	0.784615	0.785641	0.766667	0.800000
12	4	0.705507	0.691357	0.569231	0.412971	0.000000	0.725490
13	4	0.679103	0.684476	0.569231	0.412971	0.000000	0.725490
14	4	0.585710	0.534585	0.707692	0.708254	0.666667	0.739726
15	4	0.438606	0.541963	0.707692	0.708662	0.698413	0.716418
16	5	0.702770	0.664352	0.553846	0.394821	0.000000	0.712871
17	5	0.666267	0.721653	0.553846	0.394821	0.000000	0.712871
18	5	0.614998	0.544134	0.584615	0.480802	0.181818	0.721649
19	5	0.532960	0.568837	0.692308	0.668401	0.545455	0.767442

Binary Classifier (Perceived RD)

Phase 1b

	fold	Train Loss	Val Loss	Accuracy	Weighted F1	F1_0	F1_1
0	1	0.701177	0.686750	0.545455	0.547968	0.516129	0.571429
1	1	0.696874	0.606156	0.575758	0.420746	0.000000	0.730769
2	1	0.672541	0.693761	0.575758	0.420746	0.000000	0.730769
3	2	0.728308	0.693893	0.484848	0.373641	0.622222	0.190476
4	2	0.684700	0.736626	0.575758	0.420746	0.000000	0.730769
5	2	0.670451	0.611320	0.575758	0.420746	0.000000	0.730769
6	3	0.694046	0.750047	0.575758	0.420746	0.000000	0.730769
7	3	0.688924	0.611630	0.575758	0.420746	0.000000	0.730769
8	3	0.667458	0.726521	0.575758	0.420746	0.000000	0.730769
9	4	0.702659	0.668473	0.575758	0.420746	0.000000	0.730769
10	4	0.683076	0.720055	0.575758	0.420746	0.000000	0.730769
11	4	0.666856	0.715368	0.575758	0.420746	0.000000	0.730769
12	5	0.690960	0.708420	0.545455	0.385027	0.000000	0.705882
13	5	0.679503	0.748001	0.545455	0.385027	0.000000	0.705882
14	5	0.684742	0.578006	0.545455	0.385027	0.000000	0.705882
15	6	0.717765	0.698837	0.575758	0.449545	0.125000	0.720000
16	6	0.688317	0.623800	0.545455	0.385027	0.000000	0.705882
17	6	0.668637	0.804160	0.545455	0.385027	0.000000	0.705882
18	7	0.695994	0.700844	0.545455	0.385027	0.000000	0.705882
19	7	0.680631	0.614375	0.545455	0.385027	0.000000	0.705882
20	7	0.656823	0.627843	0.545455	0.385027	0.000000	0.705882
21	8	0.691792	0.680947	0.562500	0.405000	0.000000	0.720000
22	8	0.686027	0.668685	0.562500	0.405000	0.000000	0.720000
23	8	0.661941	0.652996	0.562500	0.405000	0.000000	0.720000
24	9	0.700262	0.688839	0.718750	0.712692	0.640000	0.769231
25	9	0.681517	0.680160	0.562500	0.405000	0.000000	0.720000
26	9	0.662780	0.677000	0.562500	0.405000	0.000000	0.720000
27	10	0.694381	0.681628	0.562500	0.405000	0.000000	0.720000
28	10	0.675721	0.658644	0.562500	0.405000	0.000000	0.720000
29	10	0.659534	0.634344	0.562500	0.405000	0.000000	0.720000

Aggregate summary:

	Model	LR	Epochs	Batch Size	Mean Acc	SD Acc	\
0	roberta-base	0.00002	3	32	0.564867	0.034071	

Binary Classifier (Perceived RD Intensity)

Phase 2

V18. xlsx

	Model	LR	Epochs	Batch Size	Mean Acc	SD Acc	Mean Weighted F1	SD Weighted F1	Mean F1_0	SD F1_0	Mean F1_1	SD F1_1
0	roberta-base	0.00002	4	16	0.699047	0.12376	0.639316	0.172205	0.744148	0.201905	0.443273	0.323081

RoBERTA, Google Colab
Stratified K Fold
Class Weight Loss
No Weighted Sampling

Model performance varies substantially across folds, with F1 scores for dependence ranging from 0.00 to 0.84. This instability reflects the heterogeneity and contextual nature of dependence language, which is not uniformly distributed across training samples.

Binary Classifier (Perceived RD Intensity)

6 collapsed folds

Phase 2

	fold	Train Loss	Val Loss	Accuracy	Weighted F1	F1_0	F1_1
0	1	0.698116	0.692005	0.645833	0.506857	0.784810	0.000000
1	1	0.693579	0.680626	0.729167	0.734649	0.771930	0.666667
2	1	0.658137	0.648379	0.750000	0.734244	0.823529	0.571429
3	1	0.614589	0.611516	0.770833	0.769180	0.825397	0.666667
4	2	0.693276	0.690623	0.531915	0.498542	0.450000	0.592593
5	2	0.695861	0.685286	0.659574	0.524277	0.794872	0.000000
6	2	0.662835	0.669772	0.659574	0.524277	0.794872	0.000000
7	2	0.618959	0.655516	0.680851	0.571136	0.805195	0.117647
8	3	0.691353	0.686383	0.659574	0.657108	0.652174	0.666667
9	3	0.672812	0.658346	0.851064	0.854595	0.877193	0.810811
10	3	0.656609	0.562649	0.851064	0.854595	0.877193	0.810811
11	3	0.556855	0.457815	0.851064	0.854595	0.877193	0.810811
12	4	0.696751	0.688629	0.659574	0.524277	0.794872	0.000000
13	4	0.688263	0.676231	0.595745	0.589511	0.577778	0.612245
14	4	0.668917	0.638787	0.893617	0.892745	0.920635	0.838710
15	4	0.615458	0.585379	0.851064	0.846797	0.892308	0.758621
16	5	0.696382	0.689614	0.361702	0.192154	0.000000	0.531250
17	5	0.697424	0.685593	0.638298	0.497375	0.779221	0.000000
18	5	0.668353	0.655760	0.680851	0.586786	0.800000	0.210526
19	5	0.622997	0.626733	0.659574	0.572628	0.783784	0.200000

Binary Classifier (Perceived RD Intensity)

Phase 2b

	fold	Train Loss	Val Loss	Accuracy	Weighted F1	F1_0	F1_1
0	1	0.695523	0.692741	0.666667	0.533333	0.800000	0.000000
1	1	0.691999	0.687341	0.666667	0.533333	0.800000	0.000000
2	1	0.682156	0.683033	0.666667	0.533333	0.800000	0.000000
3	1	0.685475	0.680981	0.666667	0.533333	0.800000	0.000000
4	2	0.690645	0.694552	0.333333	0.166667	0.000000	0.500000
5	2	0.694892	0.684858	0.666667	0.533333	0.800000	0.000000
6	2	0.675070	0.674885	0.666667	0.533333	0.800000	0.000000
7	2	0.659371	0.666480	0.708333	0.621083	0.820513	0.222222
8	3	0.692578	0.692811	0.333333	0.166667	0.000000	0.500000
9	3	0.687065	0.684607	0.666667	0.674074	0.733333	0.555556
10	3	0.665923	0.670175	0.666667	0.666667	0.750000	0.500000
11	3	0.648839	0.657940	0.666667	0.666667	0.750000	0.500000
12	4	0.692685	0.691922	0.666667	0.533333	0.800000	0.000000
13	4	0.693092	0.688017	0.666667	0.533333	0.800000	0.000000
14	4	0.677863	0.683047	0.708333	0.716273	0.758621	0.631579
15	4	0.673883	0.679835	0.750000	0.755245	0.769231	0.727273
16	5	0.692814	0.688916	0.375000	0.204545	0.000000	0.545455
17	5	0.698912	0.683772	0.625000	0.480769	0.769231	0.000000
18	5	0.684443	0.670754	0.833333	0.819328	0.882353	0.714286
19	5	0.684045	0.664289	0.875000	0.873340	0.903226	0.823529
20	6	0.698767	0.689897	0.375000	0.204545	0.000000	0.545455
21	6	0.695177	0.686588	0.375000	0.204545	0.000000	0.545455
22	6	0.693146	0.684246	0.708333	0.692424	0.787879	0.533333
23	6	0.681891	0.681684	0.708333	0.692424	0.787879	0.533333
24	7	0.690361	0.692726	0.652174	0.514874	0.789474	0.000000
25	7	0.694361	0.684306	0.695652	0.674369	0.787879	0.461538
26	7	0.681301	0.674723	0.782609	0.785254	0.827586	0.705882
27	7	0.671997	0.665413	0.782609	0.785254	0.827586	0.705882
28	8	0.703101	0.690232	0.347826	0.179523	0.000000	0.516129
29	8	0.691041	0.684858	0.739130	0.745151	0.769231	0.700000
30	8	0.685840	0.677959	0.695652	0.606084	0.810811	0.222222
31	8	0.670585	0.670731	0.695652	0.648221	0.800000	0.363636
32	9	0.695207	0.691346	0.652174	0.514874	0.789474	0.000000
33	9	0.688907	0.687673	0.652174	0.514874	0.789474	0.000000
34	9	0.680531	0.680685	0.652174	0.514874	0.789474	0.000000
35	9	0.664007	0.676074	0.652174	0.514874	0.789474	0.000000
36	10	0.702526	0.706063	0.347826	0.179523	0.000000	0.516129
37	10	0.698044	0.688804	0.347826	0.179523	0.000000	0.516129
38	10	0.687383	0.679808	0.739130	0.745151	0.769231	0.700000
39	10	0.669996	0.675214	0.695652	0.699356	0.758621	0.588235

Binary Classifier (Perceived RD Intensity)

Phase 3

V23. xlsx

	Model	LR	Epochs	Batch Size	Mean Acc	SD Acc	Mean Weighted F1	SD Weighted F1	Mean F1_0	SD F1_0	Mean F1_1	SD F1_1
0	roberta-base	0.00002	4	16	0.644677	0.16491	0.587997	0.227491	0.607317	0.332775	0.554565	0.187442

RoBERTa, Google Colab
Stratified K Fold
No Class Weight Loss
Weighted Sampling

Expanding the labeled dataset substantially improved model performance, enabling the classifier to reliably distinguish between low and high resource dependence. While early training epochs exhibited instability, later epochs produced consistent and balanced classification performance across folds, with F1 scores exceeding 0.70 for both classes.

Binary Classifier (Perceived RD Intensity)

Phase 3

4 collapsed folds

	fold	Train Loss	Val Loss	Accuracy	Weighted F1	F1_0	F1_1
0	1	0.696740	0.685739	0.632353	0.489931	0.774775	0.000000
1	1	0.677979	0.692894	0.367647	0.197660	0.000000	0.537634
2	1	0.627203	0.556448	0.735294	0.739938	0.763158	0.700000
3	1	0.419074	0.470369	0.735294	0.738642	0.780488	0.666667
4	2	0.694548	0.704199	0.367647	0.197660	0.000000	0.537634
5	2	0.684775	0.667640	0.705882	0.635590	0.811321	0.333333
6	2	0.684862	0.626919	0.779412	0.755143	0.848485	0.594595
7	2	0.608624	0.591515	0.808824	0.810779	0.843373	0.754717
8	3	0.697393	0.700866	0.373134	0.220217	0.045455	0.533333
9	3	0.685998	0.675704	0.701493	0.703909	0.761905	0.600000
10	3	0.574339	0.598587	0.835821	0.830786	0.879121	0.744186
11	3	0.425907	0.600454	0.761194	0.763127	0.809524	0.680000
12	4	0.696004	0.693863	0.417910	0.306874	0.170213	0.551724
13	4	0.673409	0.624674	0.701493	0.660931	0.800000	0.411765
14	4	0.589471	0.443287	0.776119	0.775024	0.827586	0.680851
15	4	0.411673	0.394154	0.820896	0.816735	0.866667	0.727273
16	5	0.705958	0.705578	0.373134	0.202790	0.000000	0.543478
17	5	0.702980	0.683265	0.582090	0.580413	0.575758	0.588235
18	5	0.640220	0.578193	0.641791	0.556831	0.769231	0.200000
19	5	0.558276	0.522041	0.776119	0.776966	0.819277	0.705882

Binary Classifier (Perceived RD Intensity)

Phase 3b

	fold	Train Loss	Val Loss	Accuracy	Weighted F1	F1_0	F1_1
0	1	0.690148	0.683119	0.647059	0.508403	0.785714	0.000000
1	1	0.703377	0.679796	0.647059	0.508403	0.785714	0.000000
2	1	0.688137	0.676917	0.647059	0.508403	0.785714	0.000000
3	1	0.682600	0.673645	0.647059	0.508403	0.785714	0.000000
4	2	0.713285	0.708862	0.647059	0.508403	0.785714	0.000000
5	2	0.692043	0.678932	0.352941	0.184143	0.000000	0.521739
6	2	0.690747	0.684209	0.823529	0.819344	0.869565	0.727273
7	2	0.679055	0.680583	0.794118	0.785982	0.851064	0.666667
8	3	0.697944	0.683625	0.647059	0.508403	0.785714	0.000000
9	3	0.688832	0.683413	0.676471	0.571946	0.800000	0.153846
10	3	0.682649	0.674691	0.794118	0.798023	0.829268	0.740741
11	3	0.655403	0.660482	0.794118	0.798023	0.829268	0.740741
12	4	0.704532	0.680401	0.647059	0.508403	0.785714	0.000000
13	4	0.691381	0.683403	0.794118	0.797887	0.810811	0.774194
14	4	0.683571	0.671869	0.500000	0.446251	0.370370	0.585366
15	4	0.659259	0.652212	0.676471	0.673389	0.666667	0.685714
16	5	0.694927	0.699156	0.382353	0.211514	0.000000	0.553191
17	5	0.686870	0.694948	0.411765	0.272262	0.090909	0.565217
18	5	0.681602	0.690529	0.529412	0.474251	0.384615	0.619048
19	5	0.635526	0.684317	0.676471	0.673950	0.666667	0.685714
20	6	0.693665	0.687308	0.617647	0.471658	0.763636	0.000000
21	6	0.693730	0.694988	0.588235	0.592525	0.611111	0.562500
22	6	0.683286	0.696844	0.558824	0.561497	0.571429	0.545455
23	6	0.648665	0.695494	0.735294	0.717862	0.808511	0.571429
24	7	0.696199	0.700030	0.382353	0.211514	0.000000	0.553191
25	7	0.688780	0.657116	0.647059	0.535014	0.777778	0.142857
26	7	0.655129	0.596267	0.764706	0.744118	0.833333	0.600000
27	7	0.603463	0.542808	0.852941	0.853871	0.878049	0.814815
28	8	0.696655	0.675200	0.363636	0.193939	0.000000	0.533333
29	8	0.688241	0.686877	0.545455	0.523426	0.482759	0.594595
30	8	0.679879	0.694956	0.757576	0.743874	0.826087	0.600000
31	8	0.663584	0.681039	0.757576	0.743874	0.826087	0.600000
32	9	0.692849	0.691472	0.545455	0.506605	0.444444	0.615385
33	9	0.684346	0.698358	0.606061	0.522638	0.745098	0.133333
34	9	0.670820	0.683456	0.696970	0.690083	0.772727	0.545455
35	9	0.648302	0.681841	0.727273	0.716883	0.800000	0.571429
36	10	0.695588	0.665388	0.636364	0.494949	0.777778	0.000000
37	10	0.685848	0.658096	0.636364	0.494949	0.777778	0.000000
38	10	0.677392	0.668798	0.696970	0.702020	0.722222	0.666667
39	10	0.635758	0.665284	0.696970	0.702020	0.722222	0.666667

Binary Classifier (Perceived RD)

Phase 4

V24. xlsx

	Model	LR	Epochs	Batch Size	Mean Acc	SD Acc	Mean Weighted F1	SD Weighted F1	Mean F1_0	SD F1_0	Mean F1_1	SD F1_1
0	roberta-base	0.00002	4	16	0.733676	0.101197	0.705246	0.151435	0.633299	0.270721	0.770781	0.054639

Model is learning the task better, but occasionally fails depending on split.

RoBERTA, Google Colab
Stratified K Fold
No Class Weight Loss
No Weighted Sampling

Binary Classifier (Perceived RD)

2 collapsed folds

Phase 4

	fold	Train Loss	Val Loss	Accuracy	Weighted F1	F1_0	F1_1
0	1	0.696145	0.662679	0.589147	0.509940	0.293333	0.710383
1	1	0.551947	0.472867	0.775194	0.769990	0.728972	0.807947
2	1	0.355285	0.553457	0.759690	0.759170	0.766917	0.752000
3	1	0.256874	0.463797	0.775194	0.775275	0.771654	0.778626
4	2	0.703262	0.666954	0.565891	0.451115	0.176471	0.705263
5	2	0.570475	0.434226	0.759690	0.750855	0.794702	0.710280
6	2	0.377443	0.323736	0.844961	0.844830	0.836066	0.852941
7	2	0.253668	0.313738	0.868217	0.868249	0.864000	0.872180
8	3	0.694697	0.678623	0.720930	0.717022	0.672727	0.756757
9	3	0.600294	0.444708	0.759690	0.754293	0.786207	0.725664
10	3	0.356214	0.414668	0.798450	0.796697	0.811594	0.783333
11	3	0.266678	0.397777	0.837209	0.837307	0.834646	0.839695
12	4	0.692051	0.690414	0.527132	0.363908	0.000000	0.690355
13	4	0.635584	0.602428	0.689922	0.681210	0.615385	0.740260
14	4	0.440954	0.460554	0.736434	0.730414	0.679245	0.776316
15	4	0.323973	0.726796	0.782946	0.781359	0.754386	0.805556
16	5	0.691395	0.675635	0.523438	0.359696	0.000000	0.687179
17	5	0.630537	0.518214	0.734375	0.729722	0.685185	0.770270
18	5	0.428796	0.444749	0.789062	0.787879	0.765217	0.808511
19	5	0.289322	0.429366	0.835938	0.835988	0.829268	0.842105

Binary Classifier (Perceived RD)

Phase 4b

	fold	Train Loss	Val Loss	Accuracy	Weighted F1	F1_0	F1_1
0	1	0.697341	0.660037	0.523077	0.359285	0.000000	0.686869
1	1	0.665796	0.704376	0.646154	0.640149	0.581818	0.693333
2	1	0.593782	0.638161	0.630769	0.584815	0.428571	0.727273
3	2	0.701654	0.696509	0.523077	0.359285	0.000000	0.686869
4	2	0.681594	0.689714	0.523077	0.359285	0.000000	0.686869
5	2	0.675792	0.718914	0.538462	0.392759	0.062500	0.693878
6	3	0.691630	0.709997	0.523077	0.359285	0.000000	0.686869
7	3	0.685749	0.665856	0.738462	0.724755	0.653061	0.790123
8	3	0.609419	0.622092	0.784615	0.779997	0.740741	0.815789
9	4	0.691681	0.666360	0.523077	0.359285	0.000000	0.686869
10	4	0.671453	0.719205	0.661538	0.609829	0.450000	0.755556
11	4	0.610314	0.547473	0.753846	0.748568	0.703704	0.789474
12	5	0.690900	0.690010	0.515625	0.350838	0.000000	0.680412
13	5	0.683107	0.675322	0.640625	0.640713	0.634921	0.646154
14	5	0.659754	0.657271	0.656250	0.655576	0.633333	0.676471
15	6	0.697329	0.686177	0.515625	0.350838	0.000000	0.680412
16	6	0.674186	0.653858	0.656250	0.604688	0.450000	0.750000
17	6	0.631298	0.595484	0.687500	0.663640	0.565217	0.756098
18	7	0.694003	0.689173	0.515625	0.350838	0.000000	0.680412
19	7	0.684011	0.679651	0.562500	0.447513	0.176471	0.702128
20	7	0.669634	0.667194	0.531250	0.384766	0.062500	0.687500
21	8	0.695480	0.691925	0.546875	0.533375	0.431373	0.623377
22	8	0.668837	0.668381	0.640625	0.633664	0.566038	0.693333
23	8	0.599538	0.636657	0.656250	0.651089	0.592593	0.702703
24	9	0.700810	0.687654	0.531250	0.368622	0.000000	0.693878
25	9	0.683268	0.673057	0.609375	0.537329	0.324324	0.725275
26	9	0.643813	0.643147	0.687500	0.666622	0.565217	0.756098
27	10	0.695481	0.684486	0.765625	0.764409	0.736842	0.788732
28	10	0.670467	0.637453	0.640625	0.586268	0.410256	0.741573
29	10	0.595351	0.561850	0.781250	0.775936	0.730769	0.815789

Binary Classifier (Perceived RD)

Phase 1

Mean ACC	SD ACC	Mean Weighted FI
0.688	0.077	0.653

Phase 4

Mean ACC	SD ACC	Mean Weighted FI
0.735	0.113	0.718

Binary Classifier (Perceived RD Intensity)

Phase 2

Mean ACC	SD ACC	Mean Weighted FI
0.759	0.073	0.726

Best Optimal Epoch within Each Fold

Epoch = 4, Fold = 5, Batch = 16

Phase 3

Mean ACC	SD ACC	Mean Weighted FI
0.795	0.0390	0.7948

Binary Classifier (Perceived RD)

Phase 1

Mean ACC	SD ACC	Mean Weighted FI
0.5621	0.0126	0.4053

Phase 4

Mean ACC	SD ACC	Mean Weighted FI
0.656	0.106	0.603

Binary Classifier (Perceived RD Intensity)

Phase 2

Mean ACC	SD ACC	Mean Weighted FI
0.7392	0.0766	0.7103

Best Optimal Epoch within Each Fold

Epoch = 3 or 4, Fold = 10, Batch = 32

Phase 3

Mean ACC	SD ACC	Mean Weighted FI
0.7410	0.0705	0.7205

Challenges & Adjustments

- **Issues:**
  - Folds collapsing to one category (Low Mean Weighted F1, Mean F1, Mean F0)
    - Sample Size
    - Weak Signal
  - Time consuming to build Class 1 (High PRD)
  - Re-classifying codes regularly in earlier stages
  - Sentence level versus paragraph level as unit of analysis

- **Changes:**
  - Class Weighted Loss
  - Weighted Sampling within Folds
  - Stratified K Fold
  - Epoch = 4
  - Learning Rate

Future Applications and Next Steps

1) **Analyze changes in perceived resource dependence (pre/post 2019)**

2) **Link to financial performance (t +1)**

3) **Link to leadership succession (background and frequency of change)**

4) **Predict on universities outside of Ontario (i.e Nova Scotia)**

Main Takeaways

- **Validate!**
  - A computer can’t do what humans can’t do

- **Need a strong signal between classes within text**

- **Need sufficient sample size in each class**